# Adi Suhit Jha

(609)-349-8914 | eMail | LinkedIn | GitHub

## Education

**University of Illinois Urbana-Champaign** — Urbana, IL
*B.S. Aerospace Engineering & B.S. Electrical Engineering, GPA:* 3.92 — *Expected May 2029*

**Princeton University** — Princeton, NJ
*Department of Physics – Dual Enrollment (High School)* — *Sept. 2024 – May 2025*

## Manuscripts and Presentations

1. **Jha, A. S.** & Padovan, A., "The harmonic resolvent operator can expose the fundamental energy-transfer pathways in fluid flows." *Journal of Fluid Mechanics* (2026). Under revision.

2. **Jha, A. S.** & Padovan, A., "Insights into energy-transfer mechanisms in fluid flows via the harmonic resolvent." *APS Division of Fluid Dynamics* (2026). Accepted.

## Research Experiences

**Reduced-Order Modeling of a Capacitative Plasma** — Undergraduate Researcher
*New Jersey Institute of Technology / Princeton Plasma Physics Laboratory* — *June 2026 – Present*

- Developing data-driven reduced-order models of capacitively coupled plasmas using Gas/NiTROM.
- Applying POD to Hermite coefficient data to initialize Petrov–Galerkin encoder and decoder bases.
- Jointly optimizing oblique projection bases and latent-space dynamics to minimize trajectory prediction error.

**Internal Energy Dynamics in Nonlinear Systems** — Undergraduate Researcher
*Padovan Group — New Jersey Institute of Technology* — *May 2025 – Present*

- Derived a framework to predict energy-transfer pathways between harmonics in nonlinear time-periodic systems.
- Developed an approximation using the harmonic resolvent that elucidates the fundamental energy transfer tensor.
- Validation against DNS of flows past a moving airfoil resulted in an energy-transfer correlation of 0.98.

## Computational Projects

**Annihilator** — Developer
*High-Performance Chess Evaluation Engine* — *Jan. 2026 – Present*

- Built a C++23 UCI-compatible chess engine using bitboards, packed move encoding, incremental make/unmake, and Zobrist hashing for low-latency search.
- Implemented alpha-beta negamax with iterative deepening, quiescence search, aspiration windows, transposition caching, pruning, and history-based move ordering.
- Designed modular evaluation and validated move generation with perft tests, benchmarking depth-6 search at 2M+ nodes/sec.

**Electromagnetic Particle-in-Cell (EPIC) Library** — Developer
*Computational Physics Library* — *Nov. 2025 – Present*

- Built a C++ particle-field simulation engine with contiguous grid storage, structured mesh indexing, and stencil-optimized interpolation kernels.
- Implemented a self-consistent PIC pipeline with CIC deposition, Boris integration, CG/multigrid Poisson solves, Gauss projection, and Yee-grid FDTD.
- Validated stability/convergence on 512x512 2D and 32x32x32 3D grids, benchmarking finite-difference kernels at nanosecond-per-cell throughput.

## Skills

**Languages:** Python, C++, MATLAB
**Tools:** NumPy, SciPy, SymPy, PyTorch, Git, Linux, MPI, OpenMP
**Methods:** Reduced-Order Modeling, Numerical PDEs, Spectral Analysis, Optimization